

LECTURE ACTIVITY 12B: HAVE A NICE TRIP – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is heavily adapted from one created by the STAT 1110 Course Coordination Team, which in turn is modified from “Introduction to Statistical Investigations,” by Tintle et. al.

Learning Objectives:

By the end of this activity, you should be able to

- 1) Identify which variable is the explanatory variable and which is response in a study involving two variables.
- 2) Identify potential confounding variables and explain how they provide an alternative explanation for the observed association between the explanatory variable and the response variable.
- 3) Identify whether confounding variables are balanced out between groups.
- 4) Explain that random assignment gives us the ability to draw cause–effect conclusions because it ensures that treatment groups have similar characteristics.
- 5) Identify whether a study uses random assignment and/or random sampling and the implications of these design decisions on the conclusions that can be drawn.

Key Terms:

- | | | |
|------------------------|------------------------|-------------------------|
| • Explanatory variable | • Confounding variable | • Experiment |
| • Response variable | • Observational study | • Randomized Experiment |

An area of research in biomechanics and gerontology concerns falls and fall-related injuries, especially for elderly people. Recent studies have focused on how individuals respond to large postural disturbances (e.g., tripping, induced slips). One question is whether subjects can be instructed to improve their recovery from such disturbances.

Suppose researchers want to compare two recovery strategies: **lowering** (quickly stepping down with front leg and then raising back leg over the object) and **elevating** (lifting front leg over the object).

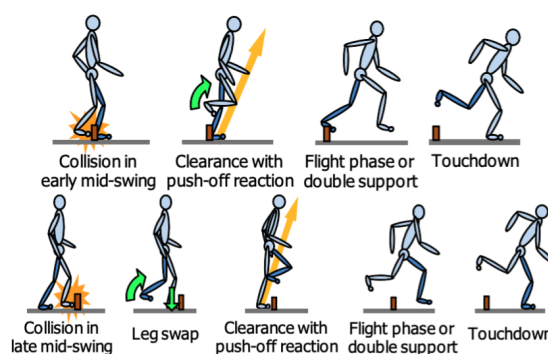


Figure 1: Illustration of the “elevating” (top) and “lowering” (bottom) strategies. Figure 2 from: Shiratori, Takaaki & Coley, Brooke & Cham, Rakie & Hodgins, Jessica. (2009). Simulating balance recovery responses to trips based on biomechanical principles. 37-46. 10.1145/1599470.1599475. **All rights retained by the original authors.**

Subjects will have first been trained on one of these two recovery strategies, and they will be asked to apply it after they feel themselves tripping. The researchers will then induce the subject to trip while walking (but harnessed for safety) using a concealed mechanical obstacle and then record whether the subject recovered from the trip or not. Suppose 24 subjects have agreed to participate in the study.

🔑 Concept Overview

When we have two variables in a study, we often consider one the **explanatory** variable and the other the **response** variable. Often the research study is looking for evidence that the explanatory variable causes changes in the response variable.

✎ Question 1

- (a) Identify the observational unit and the two variables in this study.

Solution: One observational unit is one subject. The two variables are: recovery strategy (elevating or lowering) and whether the subject recovered from the trip or not.

- (b) Which variable is the explanatory variable and which the response variable?

Solution: Recovery strategy is the explanatory variable and subject recovery is the response variable.

🔑 Concept Overview

A **confounding** variable is one whose potential effects on a response variable cannot be distinguished from those of the explanatory variable. Note that a confounding variable must be related to both the explanatory and response variables.

Suppose the 24 subjects in the sample happened to only identify as either male or female, and that they are:

- Females: Jasmine, Ka Nong, Keyaina, Mary, An, Adah, Nailah, Sassandra
- Males: Shiwoo, Jihoon, Alejandro, Daniel, Paul, Tong, Antonio, Andres, Davieon, John, Chao, Liu, Aryan, Lucas, Jabari, Rashidi

✎ Question 2

One way to design this study would be to assign the eight females to use the elevating strategy and the sixteen males to use the lowering strategy.

- (a) Would this be a reasonable strategy? What is a possible confounding variable that could affect the result of the study?

Solution: No. Assigning all females to one strategy and all males to the other would not be reasonable. Sex would be a confounding variable because women might be more likely to recover from a trip as they usually have better balance. So if all the women are assigned to the lowering strategy and that group ends up doing better, we would not know if it was because of the strategy or because of the women.

- (b) Because there will always be more potential confounding variables which are distributed unevenly between the groups being compared, identify a better method for deciding who uses which strategy.

Solution: A better method is to randomly assign subjects to the two recovery strategies in a way that half would go to one group (lowering) and the other half to the other group (elevating).

Types of Studies

🔗 Concept Overview

In an **observational study**, the groups you compare are “just there.” That is, they are defined by what you see rather than by what you do. In an **experiment**, you actively create the groups by what you choose to do to the people or experimental units. More formally, you assign the conditions to be compared. These conditions may be one or more “**treatments**” or a “**control**” (a group you do nothing to).

Note our change in terminology: We now will sometimes call the individuals in the study **experimental units** instead of observational units, because they are in an experiment instead of an observational study.

In a **randomized experiment**, you use a chance device to make the assignments. The units in an experiment are often called experimental units. **The role of the random assignment is to balance out potentially confounding variables among the explanatory variable groups, giving us the potential to draw cause-and-effect conclusions.**

Let’s explore the process of random assignment to determine whether or not it works. First, let’s focus on the sex (identify as male or not) variable. There are 16 males and 8 females. Suppose we put each person’s name on a slip, put those slips in a hat and mix them up thoroughly, and then randomly draw out 12 slips for names of people to assign to the elevating strategy.



Question 3

What proportion of the elevating group do you expect will be male? What proportion of the lowering strategy do you expect will be male? Do you think we will always get an 8/8 split (8 males in each treatment group)?

Solution: We would expect about the same proportion of males in each group, approximately $8/12 = 0.667$ (66.7%). No, we would not always get exactly 8 males in each group because random assignment produces some variability. Sometimes it would be a little higher than 8, sometimes less than 8, but one average 8 males per group.

Randomization

To repeat this random assignment a large number of times to observe the long-run behavior, we will use the [Randomizing Subjects Applet](#).

Question 4

Navigate to the applet, and click the “Randomize” button. What proportion of subjects assigned to Group 1 are male? What about for Group 2? What is the difference between these two proportions?

Solution: Answers may vary. Here are the results I got: Group 1: 0.6667, and Group 2: 0.6667. The difference (group 1 minus group 2) is 0.

You will notice that the difference in proportions of males is shown in the dot plot in the bottom graph. In this graph, each dot represents one repetition of the random assignment process where we are recording the **difference in proportions of men between the two groups**.

Question 5

Carry out the following steps in the applet:

- 1) Change the number of replications from 1 to 199 (for 200 total)
- 2) Uncheck the Animate option
- 3) Press the Randomize button. The dot plot will display the difference between the two proportions of men for each of the 200 repetitions of the random assignment process.

Include a screenshot showing the distribution with 200 repetitions.

Solution: Answers may vary. Here's my plot:

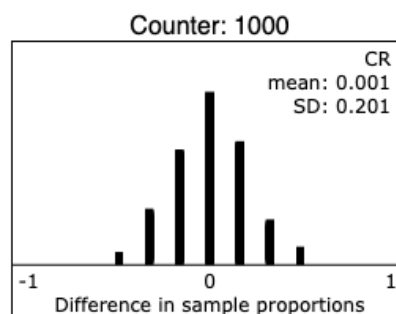


Figure 2: Screenshot from the Randomizing Subjects Applet.



Question 6

What is one dot in the distribution above? Describe the dot using the context of the study.

Solution: One dot represents one random assignment of the 24 subjects into two groups of 12 and the resulting difference in the sample proportion of males between the two groups.



Question 7

Explain why the distribution of DIFFERENCES IN PROPORTIONS OF MEN is centered around 0.

Solution: The distribution is centered around 0 because random assignment does not systematically favor one group over the other. Over many repetitions, both groups tend to receive similar proportions of males. So random assignment is balancing out the variable sex between groups.



Question 8

Does random assignment always equally distribute/balance the men and women between the two groups? Is there a tendency for there to be a similar proportion of men in the two groups? Explain based on the plot from Question (5) above.

Solution: No. Random assignment does not always perfectly balance males and females. However, the plot shows that most differences are close to 0, indicating a tendency toward similar proportions in the two groups.

🔑 Concept Overview

A comparative study is said to be **balanced** with respect to a possible confounding variable if the distribution of the variable is the same for each group in the study.

Prior research has also shown that the likelihood of falling is related to variables such as walking speed, stride rate, and height, so we would like the random assignment to distribute these variables equally between the groups as well.



Question 9

Navigate back to the applet, and carry out the following steps:

- 1) In the applet, use the pull-down menu to switch from the sex of participant variable to the height variable.
- 2) The dot plot now displays the differences in average height between Group 1 and Group 2 for these 200 repetitions.

Include a screenshot showing the distribution with 200 repetitions of DIFFERENCES IN SAMPLE MEANS for the height variable.

Solution: Answers may vary. Here's my plot:

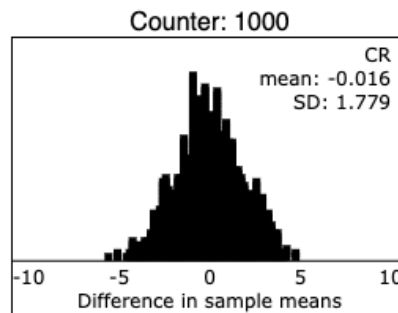


Figure 3: Screenshot from the Randomizing Subjects Applet.

 Question 10

In the long-run, does random assignment tend to equally distribute the height variable between the two groups? Explain based on the plot you obtained in the applet.

Solution: Yes. The distribution of differences in mean height is centered near 0. If a difference in sample means heights is 0 this means that the two means are the same (very similar). This shows that random assignment tends to balance height between the two groups on the long-run.

 Question 11

Suppose there is a “balance gene” that is related to people’s ability to recover from a trip.

- 1) In the applet, select the “Reveal gene?” button
- 2) Select “gene” from the pull-down menu. The applet shows you this gene information for each subject and also how the proportions with the gene differ in the two groups.

Include a screenshot showing the distribution with 200 repetitions of DIFFERENCES IN SAMPLE PROPORTIONS for the **gene** variable.

Solution: Answers may vary. Here's my plot:

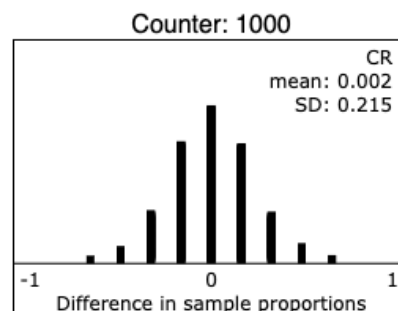


Figure 4: Screenshot from the Randomizing Subjects Applet.

 Question 12

In the long-run, does random assignment tend to equally distribute the gene variable between the two groups? Explain based on the plot you obtained in the applet.

Solution: The distribution is also centered around 0, indicating that random assignment tends to distribute the gene similarly between the two groups in the long run.

 Question 13

Suppose there was another variable like BMI that is related to people's ability to recover from a trip.

- 1) In the applet, select the "Reveal both?" button
- 2) Use pull-down menu to display the results for BMI variable.

Include a screenshot showing the distribution with 200 repetitions of DIFFERENCES IN SAMPLE MEANS for the BMI variable.

Solution: Answers may vary. Here's my plot:

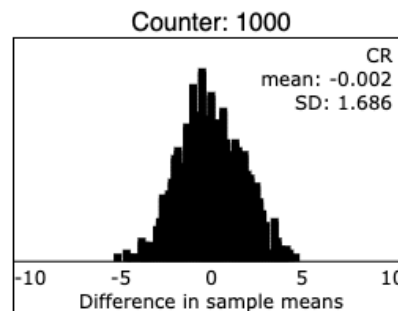


Figure 5: Screenshot from the Randomizing Subjects Applet.

 Question 14

In the long-run, does random assignment tend to equally distribute the BMI variable between the two groups? Explain based on the plot from the previous question.

Solution: Yes. The distribution is also centered near 0, showing that random assignment tends to balance BMI between the two groups over many repetitions.

 Concept Overview

Randomly assigning experimental units to groups tends to balance out all other variables between the groups. Any variables that could have an effect on the response should be equalized between the two groups and therefore should not be confounding.

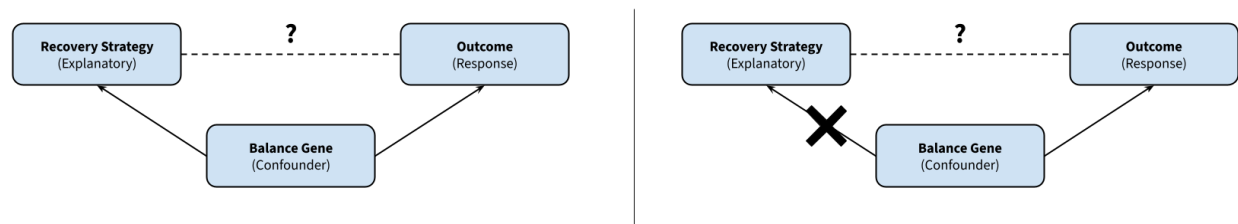


Figure 6: Illustration of the removal of confounders.

The left panel in Figure (6) shows the possible confounding effect if the study design had not been randomized. We want to know whether differences in the explanatory variable cause differences in the response (dotted arrow). However, a confounding variable would affect both the explanatory and response variables, making it impossible to separate the effects of explanatory and confounding variables on the response. The right panel shows how randomization removes the effect of the potential confounders.

Question 15

Suppose this study is conducted randomly assigning patients to each of the two strategies (lowering and elevating). This study finds a statistically discernible difference between the two treatments and finds very strong evidence that patients using the elevating strategy would show higher improvement rate than patients using the lowering strategy. What does that suggest about the potential for a cause-and-effect relationship between the explanatory and response variables? In other words, are you allowed to make a causal claim? If yes, explain why and write the causal claim; if not, explain why not.

Solution: Yes. The distribution is also centered near 0, showing that random assignment tends to balance BMI between the two groups over many repetitions.

Question 16

To what population can you apply the results of this study?

Solution: We do not have much information about the subjects in the study and we were not told that it is a random sample, so the results can be applied only to people who are similar to those in the study. Therefore, causal claims are justified but only to a specific population of subjects that are similar to the ones in the study.

		Assignment of Units to Groups			
		Random Assignment	No Random Assignment		
Selection of Units from Population	Random Sampling	A random sample is selected from one population; units are then randomly assigned to different treatment groups	Random samples are selected from existing distinct populations	⇒	<i>Inferences to population can be drawn</i>
	No Random Sampling	A group of study units is found; units are then randomly assigned to treatment groups	Collections of available units from distinct groups are examined	⇒	<i>Potential for sampling bias</i>
		⇓	⇓		
		<i>Can draw cause and effect conclusions</i>	<i>Potential for confounding variables</i>		

Table 1: Scope of Conclusions permitted depending on study design (from Ramsey and Schafer)